Source	Type	Years
Fishery	Catch biomass	1964–2025; 2026 year to date and projected annual input*
Fishery	Catch age composition	1964–2025†
Fishery	Historical trawl CPUE	1965–1976
Bottom-trawl survey	Biomass index and age-specific proportions	1982–2019, 2021–2026†
Acoustic-trawl survey	Biomass index and age-specific proportions	1994, 1996, 1997, 1999, 2000, 2002, 2004, 2006–2010, 2012, 2014, 2016, 2018, 2020, 2022, 2024, 2026†
Acoustic vessels of opportunity (AVO)	Relative backscatter index	2006–2019, 2021–2026†

**Annual catch inputs from 1991 onward use the catch estimates. Its 2026 total is 1,312,504.58 t; the within-year coverage cutoff is unspecified. †Available update observations are shown; the reference baseline ends in 2024 (fishery ages 2023); Model 26.0 uses the updated observations and selected provisional schedules, with source review retained, including the revised EBS/NBS BTS estimates. The 2020 acoustic survey used USV transects; age-specific proportions are unavailable. Provisional 2026 ATS ages use the BTS age–length key (ALK) applied to ATS population length-frequency estimates. FT-NIRS ages for the 2026 ATS samples will be available in 2027.*

FT-NIRS age estimation and survey developments are documented under age compositions, the FT-NIRS appendix, and the ATS, AVO and BTS methods sections. The assessment retains the fixed natural-mortality schedule described in Other data. Spatial catch and bycatch information helps evaluate changing distributions and guide avoidance. The assessment retains the established stock boundaries.

Model 26.0 uses observations through 2026 where available, an end-year forecast for 2026 catch, and fishery ages through 2025 (Table I1). The register identifies each source and observation year: fishery ages 2024–2025, annual survey products 2025–2026, and the scheduled 2026 ATS observation. Historical revisions, carried weights and forecast schedules have separate record types. It also distinguishes intake into the completed Model 23.0 increments, Model 23.2 bridge and selected Model 26.0 fit. Scientific acceptance remains a separate status.

4.1 Fishery data

4.1.1 Catch

Annual catch rose from about 0.175 million t in 1964 to the series maximum of 1.875 million t in 1972 (Figure 11). Catch subsequently fluctuated, fell to 0.811 and 0.810 million t in 2009 and 2010, respectively, and recovered to 1.326 million t in 2025. The 2026 records total 1.268 million t through October 17 and represent a partial year. Records before 1991 provide annual totals; later records distinguish the A-season and northwest and southeast B-season catches.

The annual assessment catch includes retained and discarded commercial pollock across gears and targets in the Bering Sea, with CDQ counted once and the separately reported Bogoslof and Aleutian Islands catches excluded. The catch update uses the catch estimates for 1991–2026, with pre-1991 model inputs retained. The revised totals are 1,311,260.74 t in 2024, 1,326,215.61 t in 2025 and 1,312,504.58 t in 2026. The 2026 estimate is the annual input selected for this assessment. Explicit Bogoslof (518) and outside-stock area codes are excluded. The 1998 records with blank area codes remain included under their Bering Sea classification. Noncommercial removals remain separately documented in Appendix C.

4.1.2 Age compositions

Fishery catch-at-age estimates for 1964–2025 are summarized in Table 5. Ages 15 and older are pooled as 15+. Where complete annual catch numbers are unavailable, we calculate numbers at age from the assessment proportions, annual fishery weight at age and catch biomass.

We estimate fishery age compositions from biological samples collected by at-sea observers and during shore-side offloading. The sampling records include fish measured for length frequencies, paired weight–length measurements,

and age estimates (Table 10; Table 11; Table 12). The processing keeps recent unsexed fish separate rather than assigning them to males or females. Within each sex and sampling stratum, we expand length frequencies to the associated catch biomass using mean weight at length. Age–length keys allocate the expanded numbers to ages; we then sum the numbers across strata and sexes and normalize them to give annual proportions (Kimura 1989; Barbeaux et al. 2005; Ianelli et al. 2024). Earlier assessments provide the pre-1991 age compositions. For the 2024–2025 update, the processing strata are January–May across Bering Sea areas, followed by June–December split into northwest (NMFS reporting areas 520 and above) and southeast (areas below 520). The 2024 assessment describes January–June and July–December groups; the current sampling groups use the calendar periods defined above. The new length and age records cover all 12 months. Unsupported age–length cells leave approximately 0.495% and 0.358% of expansion-input biomass unallocated in 2024 and 2025, respectively; these are sampling-support limits, rather than missing months or deductions from assessment catch.

The linked bootstrap represents variation among sampling units and among length and age observations within those units. Its composition precision estimates inform the selected likelihood sample sizes: 443, 578 and 530 for 2023, 2024 and 2025, respectively. They represent effective information for fitting the annual proportions, rather than numbers of fish measured or aged (Section 4.3.1). The model fits annual sex-combined proportions at ages 1–14 and 15+. Fishery age observations end in 2025.

The 2024–2025 age-estimate data are from pelagic trawl fisheries. For 2025, FT-NIRS predictions from model a00-d3 are expressed on an independent traditional microscopic-ageing (TMA) basis using 668 paired estimates (Appendix G). Within each FT-NIRS age, the paired observations define the proportions assigned to each tester age. These probabilities redistribute the catch-expanded numbers among ages while preserving their total. Source age 13 lacks paired tester observations; its conditional probabilities are the midpoint of the age-12 and age-14 rows. The 2025 source and tester ages range from 2 to 15, and the assessment fits ages 1–14 and 15+. The likelihood uses the selected input sample size of 530. Uncertainty in the conversion key and the age-13 interpolation is additional to the sampling uncertainty represented by that input weight.

The 2024 fishery age composition is expressed on the independent TMA basis using 693 paired reader–tester observations. For source age 13, the key pools TMA-minus-FT-NIRS age offsets from the same year's source ages 9–12 (34 pairs) and adds those offsets to age 13. Empirical key rows retain their observed probabilities. The conversion preserves total numbers; the selected likelihood input remains 578. Key uncertainty is additional to the sampling precision represented by that input weight.

4.1.3 Weight-at-age

Mean weight at age connects the assessment's estimates of fish numbers to catch biomass and harvest projections. Seasonal and spatial differences in fish condition and fishery distribution affect these means. Fishery sampling is therefore organized by season and area, with stratum estimates combined using the corresponding catch information. Completed-year observations provide the basis for estimating weights for the observation year and forecasting the weights needed for future catches.

For recent fishery samples, Fourier-transform near-infrared spectroscopy (FT-NIR) of otoliths provides predicted ages. Year-specific paired FT-NIR and traditional morphological ageing (TMA) observations align these predictions with the TMA age basis. Estimated abundance and biomass in each FT-NIR age bin are redistributed using the same conditional tester-age key. Mean weight at a target age is then converted biomass divided by converted abundance. For the 15+ group, biomass and abundance are pooled before taking their ratio. The conversion retains each source bin's mean weight when allocating it among tester ages. Its reported uncertainty conditions on the selected key; the supported ages and approved tail-allocation rules are documented in Appendix G.

The 2024 fishery weights use the same year-specific conversion and pooled-age rule, with the documented 100% CV for the 15+ weight observation. The 2025 weights use the same tester-age key and age-13 midpoint as the 2025 age composition. Appendix G describes these year-specific methods. The fishery age-age keys are separate from the BTS key.

The weight-model approach described in Section 14.5 of the 2024 assessment (Ianelli et al. 2024) links growth within cohorts to annual changes in condition. A smooth mean age pattern defines the typical weight increments. A cohort's weight in the previous year provides the starting point for the next age, while annual random effects allow variation in starting weight and growth. Observed mean weights and their variances inform this model, separating sampling

precision from variability among years. The fitted model supplies predictions for missing observations and forecasts for future years. Model 26.0 uses the resulting weight schedules as fixed biological inputs.

We retained the fixed 1964–1990 schedule and annual WtAgeRE observation means at ages 3–15+ for 1991–2025. Saved model predictions filled missing 2025 values at ages 13–14. Current-year predictions and forecasts omit the 2026 BTS observations from WtAgeRE: an outlier in those data had too much influence on age-3 weights in 2026 and ages 4–5 in subsequent years, through the BTS contribution and FTNIR computation. This remains a topic for future research. Age-conversion keys are treated as fixed; the 2024 fishery age-15+ weight has an assigned 100% CV.

The annual fishery weights in Table 29 retain the historical observational inputs through 2025, with fitted values for missing 2025 ages 13 and 14. The 2026 predictions and 2027–2028 forecasts use the standalone WtAgeRE fit. Ages 1–2 retain the 2024 inputs, the 1964–1990 schedule remains fixed, and the terminal age is 15+.

The 2027 and 2028 rows contain the standalone WtAgeRE fishery-weight forecasts. Age-3 weight is 0.368 kg in both years, and 15+ weight is 1.51 and 1.49 kg, respectively. Projections use the fixed 2027 fishery forecast for both catch and spawning weights.

The technical supplement documents data sources and preparation, model configurations and run identities, numerical checks, and instructions for reproducing the report.

Figure 23 shows observed fishery and BTS weights and the selected fishery-scale estimates for 2014–2028. The forecast follows the 2026 age-3 cohort from 0.368 kg to 0.482 kg at age 4 in 2027 and 0.601 kg at age 5 in 2028.

Figure 34 distinguishes the 2026 biological weight schedules from the selected 2027 catch and spawning projection weights.

4.2 Survey data

4.2.1 Bottom-trawl survey

The AFSC bottom-trawl survey provides indices of pollock abundance and population age structure, complementing the acoustic-trawl survey's sampling of the midwater component. Historical and recent sampling effort is summarized in Table 13. Annual surveys began in 1979, with standardized gear and methods from 1982 (Ianelli et al. 2024). The 2025 and 2026 observations extend the 42 survey years available for the 2024 assessment to 44 observed years during 1982–2026. The survey was absent in 2020; that year remains a gap in the observation series. The biomass estimates and assessment CVs are listed in Table 14. Availability to bottom-trawl gear varies with age and vertical distribution. Ages 2–3 are often relatively scarce in the catches because juveniles tend to occupy the water column (Monnahan et al. 2021; O'Leary et al. 2022).

Bottom temperature appears to influence the distribution of pollock available to the bottom-trawl survey. The AKsf survey-mapping example (Ianelli 2026) places annual CPUE distributions alongside the 0, 2 and 4 °C bottom-temperature contours for 2018, 2019 and 2021–2026 (Figure 12). High CPUE is concentrated along parts of the western and southern shelf, with several dense patches near the 2 °C contour in 2022, 2024 and 2026. The more extensive warm-water conditions in 2018–2019 contrast with the broader areas of cold bottom water in later panels, while the locations and extent of the pollock concentrations also change.

These patterns suggest that temperature helps structure near-bottom habitat and survey availability, with concentrations often near the margins of colder shelf water. Depth, prey, age composition and vertical movements can also contribute to the pattern. The maps show spatial associations between survey distributions and temperature. Other environmental and biological factors may also contribute. They use locally interpolated, density-dependent-corrected CPUE within survey coverage. The example's correction version and adoption status are unspecified, so its CPUE surface serves as a descriptive illustration alongside the assessment indices.

The standard EBS survey expanded northwest in 1987. Sampling in the separate NBS region has varied among years, while pollock distribution has shifted between regions. Model-based estimates integrate station observations, including zero catches, across the EBS and NBS and account for spatial and temporal variation (O'Leary et al. 2020). The biomass index uses sdmTMB and abundance at age uses tinyVAST, replacing the earlier VAST implementation. The complete selected abundance-at-age series incorporates historical revisions as well as the recent observations

(Table 15). The regional coverage, model specifications, annual covariance and remaining production-source qualifications are described in Appendix H. Estimates for unsampled years are model predictions.

The BTS length distributions show pronounced changes in the sizes represented in the survey (Figure 13). Peaks among small fish and their progression toward larger lengths illustrate cohort growth, while the relative contribution of small and large fish varies among years. These distributions complement the age compositions by showing changes in size directly.

The BTS age compositions show changes in the ages represented in the survey and the progression of cohorts through time (Figure 14). Highlighted bars follow the 1992, 2012 and 2018 year classes through successive ages. The 2012 year class progresses from age 1 in 2013 to age 10 in 2022, while the 2018 year class progresses from age 1 in 2019 to age 8 in 2026. The blank 2020 row marks the gap when the survey was absent.

The age compositions combined traditional age readings and FT-NIRS estimates. We converted the 2026 FT-NIRS estimates to traditional ages using 307 screened paired readings (Appendix G). The conversion preserved total survey abundance and pooled ages 15 and older into the 15+ group.

Spatiotemporal estimation of BTS weight at age, adopted in 2022, replaces the earlier conversion of mean length to weight (Indivero et al. 2023). The full BTS-derived weight schedules used for population biomass are presented in Table 16; the compact biological-input summary retains the recent fishery and spawning schedules and fixed biological parameters. We applied the screened 2026 age key consistently to BTS weight-at-age means and their delta-method standard errors, retaining the joint covariance treatment. We used the same key for the provisional ATS abundance and biomass-derived weights, including the weights used to predict the 2026 AVO index. We refitted the standalone random-effects weight-at-age model with the revised BTS means and standard errors under the existing equations, likelihood and source-covariance treatment, preserving fishery observations and the 2025 conversion key. These uncertainty calculations exclude empirical age-key uncertainty. The conversion, weight calculations and assessment use are described in Appendix G.

The missing 2020 survey and subsequent 2021 observation affect interpretation of the time series. Retrospective comparisons describe changes in population estimates as observations accumulate (Section 6.1.5); they do not isolate the influence of these two survey years.

4.2.2 Acoustic-trawl survey

The 2026 ATS ages are provisional: we apply the BTS age–length key (ALK) to ATS population length-frequency estimates. FT-NIRS ages from the 2026 ATS samples will be available in 2027.

Historical and acoustic-trawl sampling effort is summarized in Table 17. The 2026 collections retain their regional scope, while unavailable direct age counts remain NA.

Acoustic-trawl surveys (ATS) are typically conducted every other year and estimate the off-bottom component of the pollock stock, complementing the annual BTS index of near-bottom pollock.

Estimated pollock biomass for the EBS shelf has averaged about 3.202 million t since 1994, when the revised time series begins (Table 18). The revised estimates include the water column down to 0.5 m above the seafloor; the historical midwater index extended to 3 m above the seafloor. Low recruitment during the relatively warm early 2000s was subsequently reflected in lower survey biomass during the cold period of 2006–2012 (Honkalehto and McCarthy 2015). With growth of the strong 2012 year class, biomass exceeded 4 million t in 2014 and 2016, another warm period.

The 2026 regional estimates included the Steller sea lion conservation area (SCA) and the Northern Extension (Table 19).

Historical abundance at age

The historical series showed strong pulses of age-1 fish in 1997, 2007, 2009, 2014 and 2024 (Table 20). The provisional 2026 estimates were largest at ages 2 and 3: 3.73 and 3.96 billion fish, respectively, compared with 2.03 billion age-1 fish.

The ATS age compositions show pulses of young fish and the progression of cohorts through successive surveys (Figure 15). The highlighted 2018 year class appears at age 4 in 2022, age 6 in 2024 and age 8 in 2026. Ages 2 and 3 dominate the provisional 2026 composition. The 2026 estimates use the BTS age–length key applied to ATS population length frequencies; FT-NIRS ages from the 2026 ATS samples are expected in 2027. Blank years represent gaps in the age-composition series.

2026 survey design and results

The 2026 ATS covered the EBS shelf from approximately 60 m depth to the shelf break between 30 May and 4 August. Core-area transects were spaced at 20 nmi; selected transects extended into the Northern Extension at 40 nmi spacing. Survey timing was similar to 2022, with a longer duration in 2026 because of special projects. The area east of 170° W was surveyed from 30 May to 25 June, and the area west of 170° W from 26 June to 4 August.

The preliminary 2026 core-area estimate was 12.7 billion fish and 3.682 million t of biomass (Table 21). Biomass increased by 28% relative to the 2024 estimate of 2.875 million t (11.4 billion fish). The 2026 biomass estimate was 4% below the 2022 estimate of 3.834 million t (9.67 billion fish) and 16% above the 1994–2024 survey mean of 3.176 million t. These are survey estimates; assessment intake and adoption remain separate decisions.

Provisional 2026 ATS age compositions over supported lengths were estimated by applying the FT-NIRS-based BTS age–length key to ATS abundance-at-length (Appendix G). Midwater pollock were concentrated between the 100 m contour and the shelf break, particularly northwest of Unimak Pass and between Zhemchug and Pervenets canyons. Length distributions suggested that presumed age-1 fish (<20 cm fork length) formed dense schools on the northwestern shelf, with smaller concentrations east of the Pribilof Islands and near Unimak Pass. Presumed age-3+ fish accounted for most biomass, with 74% of their biomass west of 170° W.

Provisional abundance at age from the 2026 BTS keys

We estimated provisional ages by matching ATS lengths to the 2026 BTS FT-NIRS age–length key, then converting those ages to traditional ages using paired readings. Ages 2 and 3 dominated the resulting distribution (Table 22; Figure 16). Fish beyond the key's length range represented less than 0.001% of core-area abundance. The technical appendix describes the keys, calibration and allocation assumptions.

These ATS age estimates are provisional and will be updated with ATS age-composition data in 2027. Assessment comparison and final model selection remain separate from this derivation.

Biomass and mean weight at age

We applied the same age keys to biomass and divided biomass by abundance to estimate mean weights at age. The biomass allocation and mean weights are presented in Table 23, Table 24 and Figure 17. These estimates use the same age compositions.

Survey uncertainty and assessment treatment

We estimated ATS biomass sampling error with the established one-dimensional geostatistical method, which accounted for spatial patterns along survey transects (Petitgas 1993; Walline 2007; Williamson and Traynor 1996). The 2026 sampling error was 3.4% of estimated biomass, compared with a historical average of 4.4%. Complete core-area coverage at 20 nmi transect spacing likely contributed to the lower error.

For assessment use, we increased the survey errors to allow for additional uncertainty, including target strength and trawl selectivity. We scaled the assessment coefficients of variation to average 20% over 1994–2026 and assigned a value of 45% to 2000 (Table 18). We retained the established geostatistical method because the approach described by Urmy et al. (2025) was unavailable for production use in 2026.

The assessment used 20 biomass observations through 2026. Table 25 describes the separate treatment of survey age compositions.

Spatial and historical patterns

The Northern Extension contained an additional 0.078 million t beyond the 3.682 million t core-area estimate (Table 19).

The ATS index used in the assessment covered the water column down to 0.5 m above the seafloor, while the historical heatmap showed biomass from near the surface to 3 m above the bottom (Figure 18). Biomass was low during 2008–2012, especially in 2008–2009, and increased in 2014–2016. The maps also showed shifts in the strength and location of concentrations along the western and southern shelf.

Biomass time series and observation errors

The series contained 20 survey estimates between 1994 and 2026. We plotted approximate 95% sampling intervals around each biomass estimate using the survey standard errors (Figure 19; Table 18). Gaps in the annual series marked years without a survey estimate. These intervals described survey sampling variability; the assessment used larger errors to allow for additional uncertainty.

Biomass varied strongly along the survey tracks. Concentrations were highest on parts of the western and southern shelf and generally lower in the far north (Figure 20). The map included the Northern Extension and showed GMT contours of mean biomass per sampling interval (tons per EDSU). The map uses approximately 10-nautical-mile cell means, with the smoothed surface displayed within 30 km of sampled cell centers.

The western region contained the greatest biomass and abundance, followed by the eastern region and the SCA (Table 19).

4.2.3 Acoustic vessel-of-opportunity index

The acoustic vessel-of-opportunity (AVO) index provides a relative-abundance observation for age-1 and older midwater pollock within the U.S. EEZ (Honkalehto et al. 2011; Lauffenburger et al. 2025). The 2026 observations were collected at 38 kHz from 29 May to 5 August aboard the summer bottom-trawl charter vessels. The index covers the water column from 3 m above the seabed to 16 m below the surface.

Annual index

Relative AVO backscatter is lowest at 0.299 in 2009 and highest at 1.55 in 2015 (Table 26; Figure 21). It is 0.863 in 2024, 1.07 in 2025 and 1.12 in 2026, with sampling CVs of 2.72%, 5.26% and 4.55%, respectively. The 2025 and 2026 sampling confidence intervals overlap. The display series covers 2009–2026, excluding 2020, when no bottom-trawl survey took place. Each annual area-integrated estimate is divided by the mean of the 17 estimates to give a mean-one series. The table also retains total pollock backscatter, in millions of square metres, with sampling errors and scaled confidence intervals. This is an acoustic backscattering cross section, rather than geographic area (Lauffenburger et al. 2025).

For the nine paired survey years in the 2009–2026 display period, AVO total pollock backscatter and revised ATS biomass have a squared Pearson correlation of 0.89. This association is descriptive: AVO covers 3 m above the seabed to 16 m below the surface, while the revised ATS index extends from 0.5 m above the seabed to 16 m below the surface. The vertical domains differ. The 2026 AVO report gives 0.90 using an ATS 2026 estimate of 3.62 million t; the comparison here uses the revised ATS estimate of 3.682 million t (AFSC-RACE MACE 2026).

Spatial distribution and sampling coverage

Strong backscatter occurs along western EBS tracks in the recent maps, with weaker returns interspersed along central and southeastern tracks (Figure 22). The annual panels use a common map extent and color scale while showing differences in track coverage.

The longer series also contains alternating strong and weak returns along the sampled shelf (Figure 22). The year 2020 is omitted because AVO observations are unavailable. Survey coverage and processing vary among years. Inferences about population redistribution require comparable sampling and integration across years.

The number of observed survey intervals ranges from 5,557 in 2009 to 12,229 in 2024 (Table 27). Recorded-zero shares fall from 56.3% in 2009 to 7.8% in 2015, vary subsequently, and are 14.9%, 5.0% and 14.2% in 2024–2026, respectively.

The maps summarize observed pollock backscatter in approximately 20-nautical-mile cells, including recorded zeros. GMT gridding and contours show the mean backscatter in m^2 nmi^2 on a common color scale. Surfaces extend up to 45 km from sampled cell centers; more distant areas remain masked. The annual AVO index integrates backscatter

over the survey area and is reported in million m². Survey coverage and processing vary among years, so comparisons of spatial patterns depend on comparable sampling.

Assessment use and observation uncertainty

The AVO assessment index is 2.01 million m² in 2024, 2.50 million m² in 2025 and 2.60 million m² in 2026 (Table 28). Its 20 observations cover 2006–2026, excluding 2020. The assessment uses the newly processed 2011 and 2013 estimates and their sampling errors (AFSC-RACE MACE 2026). The 2006–2008 estimates retain the legacy rescaling described in the 2024 assessment (Ianelli et al. 2024). Mean-one display normalization remains separate from the assessment inputs in million square metres.

Survey sampling errors describe one-dimensional geostatistical variability. The assessment applies additional observation-error inflation to an arithmetic mean CV of 23% across its 20 observations. The two newly processed years use the existing modern-year error inflation rule, followed by common recalibration of all 20 CVs to that mean. Assessment errors differ from sampling errors and from the ATS mean assessment CV of 20%.

For the Normal index likelihood, arithmetic observation SEs equal the retained assessment index multiplied by its selected CV. For the display series, let It denote the original index, I¯ its 17-year mean and Rt=It/I¯. $R_t=I_t/\bar I$. The same constant rescales its index, SEs and confidence limits, leaving the CV unchanged:

$$\mathrm{SE}(R_t)=R_t\,\mathrm{CV}_t,\qquad [L_t,U_t]=\frac{I_t\pm1.96\,I_t\mathrm{CV}_t}{\bar I}.$$

The source 95% interval is represented by the half-width 1.96ItCVt. $1.96\,I_t\mathrm{CV}_t$, checked for every observed year. The displayed intervals describe survey sampling uncertainty with normalization treated as a change of units; assessment error inflation is a separate model input.

4.3 Other data

Model 26.0 fitted the 12 historical CPUE observations from 1965–1976 using fishery selectivity and fishery weights at age. We retained the observations and errors from the reference fit through 2024.

Fixed biological assumptions

We retained the 2024 biological assumptions (Ianelli et al. 2024). Natural mortality was 0.9 yr□¹ at age 1, 0.45 yr□¹ at age 2 and 0.3 yr□¹ at ages 3–15+. Maturity at ages 1–15+ was 0, .008, .289, .641, .842, .901, .947, .963, .970, 1, 1, 1, 1, 1, 1. The model calculated April spawning biomass one quarter-year after the start of the year and applied a female fraction of 0.5 once to sex-combined abundance. We used a 15-by-15 identity matrix for ageing error, matching the archived ADMB configuration. This likelihood assumption treated ages as exact after data preparation, which used FT-NIRS age conversion and age–length keys.

All weight schedules covered ages 1–15+ for the combined sexes in kg per fish. The model used fishery weights to calculate catch biomass, BTS weights to calculate population biomass and A-season weights to calculate spawning biomass.

Population and spawning weights

Population weights use BTS observations through 2026. Three historical values retain their previous treatment; details are in the technical supplement.

For spawning biomass, we replaced the older weights at ages 3–15+ with A-season fitted values for 2024–2025 and projections for 2026. Within the common-year comparison, this update changed the 13 values at ages 3–15+ in 2024. We retained baseline weights at ages 1–2; age-2 maturity was .008.

ATS and AVO weights

We retained each historical schedule through 2024 and carried its weights forward to 2025. After BTS-ALK and conditional TMA preparation, we calculated provisional 2026 ATS weights by dividing biomass by the corresponding abundance. For age 15+, we pooled biomass and abundance before division. Appendix G documented the supported lengths and unallocated tail. We used the 2026 ATS vector as a proxy for AVO weights, although the historical schedules differed.

Section 5.2 described the rationale for changing weight estimation. Weight-schedule uncertainty remained separate from the assessment's conditional parameter uncertainty. Native forecasts repeated the 2026 weight vectors. The proposed Tier 3 handoff used its documented fishery forecast and carried-forward weights. The projection inputs and catch recommendations are presented in the projection and harvest-advice sections.

4.3.1 Age-composition input sample sizes

The full-data Model 23.0 fit uses fishery composition N=443, 578 and 530 for 2023, 2024 and 2025, respectively (Table 25). These are the rounded continuity-scaled harmonic estimates from the corrected sampler bootstrap, applied in the completed full-data Model 23.0 fit at the author's direction. The bootstrap preserves linked sampling units within fishery strata and uses 10,000 replicates per year. ATS 2026 uses N=94, and BTS retains the selected 2026 N, shown as 131. Fishery N is fixed at 10 in each year from 1964–1977. The three composition-likelihood multipliers are 1.

Model 26.0 fits age compositions with a multinomial model. The baseline retains likelihood sample sizes from the bridge workbook. These likelihood sample sizes differ from counts of fish sampled and from diagnostic effective sample sizes. For every fishery or survey year, the final assessment must provide the raw sampling information, starting sample size, adjustment method, and final sample size used in the likelihood. The 2020 biomass estimate remains a separate fitted survey observation.

One-step-ahead (OSA) residuals compare an observation with its conditional prediction after preceding age categories have been considered. The OSA and Francis comparisons document the selected weights and observed-data residual patterns in Section 6.1.6.

5 Analytic approach

5.1 Main and comparison models

Model 26.0 (Rceattle) is the adopted model for Tier 3 calculations and specifications. Models 23.0 and 23.2 provide comparisons with the previous assessment and the transition in model structure. Other sensitivities separate the early Foreign fishery (1964–1990) from the later domestic fishery (1991–2026) and vary the penalty on declining selectivity.

Model 26.1.2 is the Early-late fishery split sensitivity. The fits and retrospective comparisons hold the process standard deviations fixed at the values estimated by its marginal model. Model 26.0 uncertainty and retrospective diagnostics are described below.

5.1.1 Model 23.0 and impact of new data

As in past years, we incrementally introduced new data relative to the previous assessment while retaining the Model 23.0 formulation. We began with total catch through 2026, incorporating the revised catch estimates for 1991–2026 (2026: 1,312,504.58 t) and extending the model endpoint and parameter and projection horizons to accommodate the added years. At this first step, fishery age compositions and survey observations retained their historical coverage. We then updated weight at age and A-season spawning weights, followed by fishery catch-at-age data and their information weights.

The survey updates followed in sequence: the bottom-trawl survey (BTS), the acoustic-trawl survey (ATS), and the acoustic vessels of opportunity (AVO) index. BTS retained the 2020 survey gap and annual information weights, including the real-valued 2026 input. ATS incorporated the selected age-1 rule and provisional 2026 age information, and AVO incorporated its annual observations and assessment-error recalibration. Historical revisions and uncertainty recalibration were included with each survey's update (Section 4.2.2; Section 4.2.3).

Each step includes all preceding updates, so its estimated effect depends on the order in which the data were added. The resulting 2026 estimates are summarized in Table 30, and the cumulative annual point estimates are shown in Figure 24. These comparisons use matched definitions of population-weight biomass, female spawning biomass, age-1 recruitment and maximum realized fishing mortality across ages. Model equations are given in the model-equations appendix.

The saved 2027 spawning-biomass and young-abundance projections from the cumulative input-update analyses are compared in Figure 25. These diagnostic projections retain the earlier 2026 catch estimate of 1,310,000 t; the 2026

estimates above use the revised catch inputs. The display starts with Catch + 2026 endpoint, followed by + Fishery age data and the subsequent survey updates. The intervening weight-at-age and A-season spawning-weight updates remain incorporated in the fishery-age and later fits. Abundance at ages 1–3 is measured at the start of 2027, before removals; spawning biomass includes mortality to spawning time. Projection settings are detailed in the technical supplement. Management projections retain their separate catch, selectivity, weight and recruitment assumptions (Section 7). The selected fishery information weights remain 443, 578 and 530 for 2023–2025. The two recent fishery compositions use their respective TMA conversions (Section 4).

The input sequence and configuration comparison answer different questions. The former retains the Model 23.0 formulation while adding observations and updating schedules. The latter compares formulations on matched observations and biology (Table 31). The historical 2024 advice is retained for reference.

Approximate distributions compare the change in spawning biomass from 2026 to 2027 (Figure 26) and the historical change in total abundance at ages 1–4 from 2025 to 2026 (Figure 27). They show the same selected cumulative sequence, followed by Models 23.2 and 26.0. Their input and projection settings are documented in the technical supplement. Each distribution retains its matching fitted-parameter covariance between years and, for abundance, among ages. The normal approximations are conditional on the model and biological and projection assumptions. They describe local estimation uncertainty and are separate from posterior inference.

Among the additions in the saved 2027 projection comparison, ATS produced the largest increase in female spawning biomass: from 2.685 to 2.975 million t, an increase of about 290,000 t (10.8%) relative to the preceding BTS step. AVO increased projected spawning biomass a further 61,000 t (2.0%), to 3.036 million t. The effects on estimated 2026 spawning biomass differed: fishery age data produced the largest increase, about 213,000 t (8.4%); ATS reduced it by about 73,000 t (2.6%), and AVO increased it by about 47,000 t (1.7%). These are successive changes within the specified sequence, with each survey contrast including its historical revisions and error calibration.

5.1.2 Model 23.2 fitted to the 2026 assessment data

Model 23.2 is a structural bridge between the ADMB continuity model and the Rceattle configuration. Its annual spawning-biomass and recruitment trajectories retain their documented scope. The established zero-variance management quantities are excluded from management comparisons. Current configuration effects use verified saved population outputs and their stated units. The comparison of Models 23.0 and 23.2 includes matching local asymptotic uncertainty ranges (Figure 28). Biomass uses common population weights, and fishing mortality is the maximum realized rate across ages. Positive annual quantities use the stated first-order log-scale intervals.

5.1.3 Model 26.0 fitted to the 2026 assessment data

Model 26.0 uses one fishery and fixed weight-at-age schedules. The updated base model fit retains its native working-likelihood, parameter bounds, selectivity forms, priors and fixed biological assumptions. We check outputs against the written inputs and saved native quantities. Population-weight biomass, female spawning biomass, age-1 recruitment and maximum fishing mortality retain distinct definitions. Their annual estimates and matching local asymptotic uncertainty are compared with Model 23.2 in Figure 29. These ranges use each model’s own covariance; each interval describes uncertainty around its fitted estimate.